

Decision and Social Choice

PE3101P — Lecture 7: Harsanyi's aggregation theorem

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Mid-Course Feedback: The Second Half

- ▶ Weeks 1–2 were too slow and weeks 4–5 too squeezed. The second half will be slower and better balanced.
- ▶ Slightly less mathematical, though still somewhat technical.
- ▶ Three topics, two lectures each:
 - ▶ **Weeks 7–8:** Harsanyi; egalitarianism and prioritarianism.
 - ▶ **Weeks 9–10:** social choice theory and a little game theory; Sen's liberal paradox.
 - ▶ **Weeks 11–12:** voting theory; the alleged irrationality of voting.

Logistics

- ▶ **Readings** are now optional.
- ▶ **Slides**: two versions from now on, with and without transitions.
- ▶ **Assessments**: broadly the same. Test 2 may have more text-based questions, but with objectively correct answers: no essays.
- ▶ **Optional make-up tutorial** (the Friday tutorial falls on Well-Being Day): Thursday 8 October, 10:00–12:00, AS3-0305.

Today

- 1 Social Choice
- 2 Aggregation
- 3 The Proof
- 4 Utility and Wellbeing
- 5 Objections
- 6 More on the Premises

Choosing for One, Choosing for Many

- ▶ A person's preferences tell us how they will decide for themselves.
- ▶ If a utility function represents them, they prefer whatever has higher expected utility.
- ▶ But that doesn't tell us how to choose for several people, when the decision affects them differently.
- ▶ They can disagree, so there might be no option that is "best for everyone".
- ▶ We need a way to balance people's interests against each other. Harsanyi's aggregation theorem gives an argument for one way of doing that.

Two Kinds of Disagreement

- ▶ Two of us both want higher economic growth.
- ▶ The liberal thinks it will come from lower taxes. The social democrat thinks it will come from raising taxes to invest more in education.
- ▶ That disagreement comes from different credences about what will happen. It's not the kind we're interested in here.
- ▶ We're interested in cases where people are affected differently: one is made better off, the other worse off.
- ▶ So we judge everything from *one* set of credences: the same probabilities for everyone.

A Bigger Benefit, or a Smaller One?

	Option 1	Option 2
Alex	8	2
Darius	3	5

Wellbeing levels.

- ▶ Option 1 is better for Alex, by 6. Option 2 is better for Darius, by 2.
- ▶ One natural answer: benefit someone by more, rather than someone else by less. Choose Option 1.

Question

Could we get numbers like these from their preferences alone, using their utility functions? What might go wrong?

Stretching a Utility Function

If u represents someone's preferences,
so does $au + c$, for any $a > 0$.

	Option 1	Option 2
Alex's utility	8	2
Darius's utility	30	50

Stretch Darius's utility function by a factor of 10. It represents exactly the same preferences, but now his difference is 20.

Preferences Don't Fix the Scale

- ▶ The difference in *utility* depends on how much each function is stretched. Nothing stops us stretching Alex's, or shrinking Darius's.
- ▶ So if all we look at is their preferences, there's no way to tell which option makes more of a difference.
- ▶ We have to bring in some other information. That's the main reason we talk about wellbeing.

We Need Comparisons of Wellbeing

	Option 1	Option 2
Alex	100	0
Darius	4	5

Wellbeing levels.

- ▶ A massive difference for Alex, a tiny one for Darius, and no reason to favour Darius.
- ▶ Overall, the best person to benefit is Alex. But unless we can compare their wellbeing, we can't say that.
- ▶ Preferences are enough for decision theory, but arguably not to decide what's best overall.
- ▶ (Deciding from preferences alone is still sometimes useful: that's voting theory, later in the semester.)

Betterness, Not Preference

Not: X is at least as *preferred* by i as Y .

But: X is at least as *good for* i as Y . Written $X \succsim_i Y$.

- ▶ Why: we need facts about wellbeing, and “good for i ” is about wellbeing, not preference. (On its own this won’t fix the scale: more after the proof.)
- ▶ $X \sim_i Y$ means equally good for i , and $X \succ_i Y$ means better for i .
- ▶ It ranks prospects, not just outcomes: it takes account of each person’s risks and benefits.
- ▶ An outcome counts as a prospect: the one that gives that outcome for sure.

The Social Welfare Ordering

- ▶ Like a betterness ordering for the whole of society: everyone affected by the decision. It ranks prospects too.
- ▶ $X \succsim Y$: X is at least as good as Y , overall.
 $X \sim Y$: equally good. $X \succ Y$: better.
- ▶ It could be read as:
 - ▶ what society *should* choose;
 - ▶ what society *will* choose;
 - ▶ which outcome *ought to be chosen*, morally;
 - ▶ which outcome would be *morally best*.
- ▶ I'll mostly have the last in mind: which outcome is best, when the choice involves trade-offs between people.

Harsanyi's Aggregation Theorem

- ▶ Starts with some really plausible-looking principles.
- ▶ Derives a conclusion that is far from impossible to believe, but definitely controversial:

Utilitarianism

The best outcome is the one with the greatest sum of wellbeing*:

$$X \succsim Y \quad \text{iff} \quad w_1(X) + \cdots + w_n(X) \geq w_1(Y) + \cdots + w_n(Y),$$

where $w_i(X)$ is person i 's wellbeing in outcome X .

*Strictly, a sum of utilities chosen to match wellbeing: more after the proof.

Same Total, Equally Good

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

- ▶ Utilitarianism: these are equally good.
- ▶ Is giving everything to Alex really equally as good as sharing everything out equally?!
- ▶ The proof will get from UNEQUAL to EQUAL one step at a time. Watch for the step you would resist.

Anonymity

Anonymity

If outcome Y can be obtained from outcome X by rearranging who gets which wellbeing level, then $X \sim Y$.

- ▶ Writing outcomes as (Alex, Darius): $(10, 0) \sim (0, 10)$.
- ▶ Alex and Darius matter equally, and what's at stake for them is equal. So we should be indifferent about who gets what.
- ▶ With Philip too, as (Alex, Darius, Philip):
 $(1, 2, 3) \sim (2, 3, 1)$. And so on, for any number of people.

Statewise Dominance

Statewise Dominance

If prospect X is at least as good as prospect Y in every state of nature, then $X \succsim Y$. If X is also better than Y in some state of nature, then $X \succ Y$.

- From here on, the premises are about prospects.

	Disaster	No disaster
Prepare	less damage	as normal
Don't prepare	more damage	as normal

Preparing costs nothing. If the disaster happens, preparing was better; if not, it's equally good either way.

Ex Ante Pareto

Ex Ante Pareto

If prospect X is at least as good for everyone as prospect Y , then $X \succsim Y$. If X is also better for someone than Y , then $X \succ Y$.

- ▶ “Ex ante”: it applies to prospects, not just to outcomes.
- ▶ Choosing for many people still involves risk. Think of economic policy.
- ▶ “Better for everyone” doesn’t mean guaranteed to turn out better for everyone. It means a better balance of risks and benefits for each person.

Expected Utility: What Is Better for Each Person

- ▶ How do we decide what's better for each person? As in the first half:

Expected Utility

Each person's betterness relation satisfies the axioms of expected utility theory. So it can be represented by a utility function, in the expected-utility sense.

- ▶ The numbers we assign are still utilities: just utility functions constrained by the facts about wellbeing.
- ▶ What exactly that means: after the proof.

Transitivity

Transitivity

If $X \succsim Y$ and $Y \succsim Z$, then $X \succsim Z$.

- This is what chains the steps of the proof together.

The Premises

- ▶ Anonymity
- ▶ Statewise Dominance
- ▶ Ex Ante Pareto
- ▶ Expected Utility (for each person's betterness)
- ▶ Transitivity

Also assumed: every distribution of wellbeing, and every lottery over them with finitely many outcomes, is possible.

They all look fairly innocuous. But together, they imply utilitarianism.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Equally good in every state (Heads is unchanged).

By Statewise Dominance, both ways round:

UNEQUAL $\sim$ EQUAL CHANCES.

A Simple Case (2): EQUAL CHANCES and EQUAL

EQUAL

	Heads	Tails
Alex	5	5
Darius	5	5

By Expected Utility, equally good for each person.

By Ex Ante Pareto, both ways round:

EQUAL CHANCES $\sim$ EQUAL.

Putting It Together

UNEQUAL $\sim$ EQUAL CHANCES	Anonymity, Statewise Dominance
EQUAL CHANCES $\sim$ EQUAL	Expected Utility, Ex Ante Pareto
UNEQUAL $\sim$ EQUAL	Transitivity

(10, 0) is exactly as good as (5, 5). They have the same total.

That's the heart of the argument. The general case makes the same two moves with any number of people, then uses them to rank any two outcomes.

The Equalisation of an Outcome

Equalisation

The equalisation of outcome X , written $E(X)$: everyone gets the average wellbeing in X , for sure.

- ▶ Three people, Alex, Darius and Philip: $X = (1, 2, 3)$.
- ▶ The average is 2, so $E(X) = (2, 2, 2)$.
- ▶ For $X = (10, 0)$: $E(X) = (5, 5)$, which is EQUAL.

The Lottery-Equalisation of an Outcome

Lottery-Equalisation

A lottery-equalisation of X , written $LE(X)$ (any one will do): n equally likely states, one per person. In each state, the outcome is a rearrangement of X , and each person takes each place in X in exactly one state.

$LE(X)$, for $X = (1, 2, 3)$

	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

For $X = (10, 0)$: $LE(X)$ is EQUAL CHANCES.

Step 1: $X \sim \text{LE}(X)$

LE(X)			
	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

So the two prospects are equally good in every state.
By Statewise Dominance, both ways round: $X \sim \text{LE}(X)$.

Step 2: $LE(X) \sim E(X)$

E(X)			
	State 1	State 2	State 3
Alex	2	2	2
Darius	2	2	2
Philip	2	2	2

Each state has probability $\frac{1}{3}$.

By Expected Utility, equally good for each person.
By Ex Ante Pareto, both ways round: $LE(X) \sim E(X)$.

Step 3: $X \sim E(X)$

$X \sim LE(X)$ Step 1 (Anonymity, Statewise Dominance)

$LE(X) \sim E(X)$ Step 2 (Expected Utility, Ex Ante Pareto)

$X \sim E(X)$ Transitivity

Every outcome is equally as good as its equalisation.

(In the simple case: $(10, 0) \sim (5, 5)$.)

Step 4: Utilitarianism

- ▶ Compare any two outcomes X and Y .
- ▶ If X has the greater total wellbeing, then in its equalisation each person will be better off: everyone is better off in $E(X)$ than in $E(Y)$.
- ▶ By Ex Ante Pareto, $E(X) \succ E(Y)$.
- ▶ So $X \sim E(X) \succ E(Y) \sim Y$. By Transitivity, $X \succ Y$.
- ▶ If the totals are equal, $E(X)$ and $E(Y)$ are the same outcome, so $X \sim Y$.

$X \succsim Y$ iff X has at least as great total wellbeing as Y .

The Harsanyi–Sen–Weymark Debate

- ▶ The debate: what does the aggregation theorem show?
- ▶ The objection: it shows that overall good is a sum of utilities. Nothing connects utility with wellbeing. So it doesn't show utilitarianism.
- ▶ So does the theorem really show utilitarianism? How might a utilitarian reply to the objection?

The Numbers Are Utilities

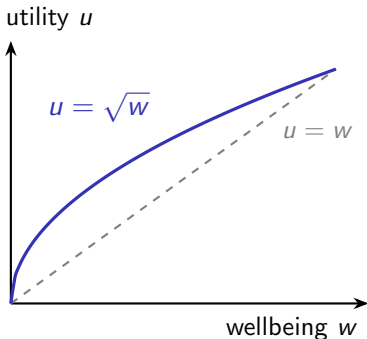
- ▶ In the proof, we treated the numbers as wellbeing levels. Strictly, they are utilities.
- ▶ By Expected Utility, each person's betterness relation has a utility function: numbers whose expectation says what's better for that person.
- ▶ But only up to stretching and shifting: the problem we started with.
- ▶ So which utility functions are we using? And how do they relate to wellbeing?

Utility Isn't Defined as Wellbeing

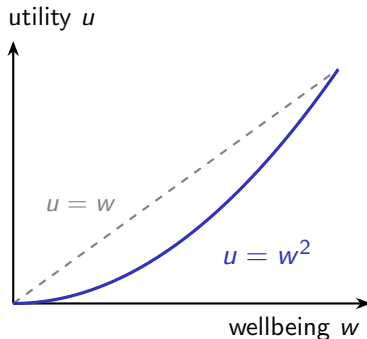
- ▶ Utility is a technical notion. It's defined by each person's betterness over prospects: whatever numbers have the right expectations.
- ▶ Wellbeing is how well off someone is.
- ▶ It's tempting to think utility just *is* the amount of wellbeing. But nothing in the definition says so.
- ▶ The two can come apart, by a curve.

Utility as a Function of Wellbeing

Two ways utility could curve against wellbeing.
The dashed line, $u = w$, is for comparison.



Concave:
risk averse about wellbeing



Convex:
risk seeking about wellbeing

Risk Attitudes About Wellbeing

- ▶ Suppose Darius's utility is $\sqrt{\text{wellbeing}}$.

Wellbeing	0	1	4	9
Darius's utility	0	1	2	3

- ▶ 4 for sure: utility 2. A 50–50 chance of 1 or 9: expected utility $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 3 = 2$.
- ▶ So they're equally good for Darius, though the gamble gives more wellbeing on average (5). His betterness is risk averse about wellbeing.
- ▶ But it's never risk averse about utility: by definition, expected utility is what's better for him.

Same Scale

- ▶ To compare people, we use the facts about wellbeing to fit their utility functions to one another:

Same Scale

When two people get the same number, they are equally well off.

- ▶ This needs only comparisons of levels (“equally well off”), not amounts of wellbeing (“twice as well off”).
- ▶ With Same Scale, the same utility means the same wellbeing, across people.
- ▶ But not necessarily twice the utility, twice the wellbeing.

Same Scale Needs the Same Risk Attitude

- ▶ Can we always meet Same Scale? Only if everyone's betterness has the same attitude to risk about wellbeing: the same curve for everyone.
- ▶ Suppose Alex's betterness is risk neutral about wellbeing, and Darius's is risk averse:

Wellbeing	0	1	4	9
Alex's utility = wellbeing	0	1	4	9
Darius's utility = $\sqrt{\text{wellbeing}}$	0	1	2	3

Match them at 0 and 1, and they come apart at 4 and 9. No stretching or shifting fixes that.

Why Betterness Shares a Risk Attitude

- ▶ So is it plausible that everyone's betterness has the same risk attitude?
- ▶ For preferences, maybe not. Unless rationality requires a particular risk attitude, a range might be rationally permissible, and people's preferences needn't share one.
- ▶ But betterness tacitly uses the correct risk attitude. We're judging the best balance of risk for that person, which doesn't seem to depend so much on the attitudes of the people in question.
- ▶ What's good for you is good for you, regardless of what you prefer (arguably, anyway).

What the Theorem Really Says

- ▶ For each person: the numbers whose *expectation* says what's good for them.
- ▶ The theorem: the *sum* of those same numbers says what's good overall.
- ▶ “The greatest sum of wellbeing” is exactly right only if betterness is risk neutral about wellbeing: utility a straight line in wellbeing.
- ▶ Otherwise, it's the greatest sum of utilities. That's the asterisk.

VNM Utilitarianism

- ▶ If the premises are true, the theorem's conclusion is not necessarily utilitarianism.
- ▶ Call it *VNM utilitarianism* (VNM: von Neumann–Morgenstern): overall betterness follows the total of utilities, of the specific kind linked to wellbeing in the right way.
- ▶ Is that utility wellbeing? One response: before decision theory, we had no coherent notion of quantities of wellbeing; decision theory gives it content, and utility is wellbeing. That may or may not be true.
- ▶ So the debate is about whether VNM utilitarianism is utilitarianism.

Can We Reject the Conclusion?

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

- ▶ Back to the table the proof started from. Utilitarianism: these are equally good.
- ▶ The apparent unfairness: we're indifferent between sharing equally between Alex and Darius, and giving everything to Alex. Intuitively, that's clearly unfair.
- ▶ Two fairness-based objections: one to each move in the proof, and the premises behind it.

UNEQUAL $\sim$ EQUAL CHANCES Statewise Dominance (and Anonymity)

EQUAL CHANCES $\sim$ EQUAL Ex Ante Pareto (and Expected Utility)

Diamond's Objection

	Heads	Tails
UNEQUAL	(10, 0)	(10, 0)
EQUAL CHANCES	(10, 0)	(0, 10)

- ▶ The coin toss seems better, because it's fairer: Darius at least gets a chance.
- ▶ But it is no better in either state. If Heads, the outcome is the same. If Tails, $(0, 10) \sim (10, 0)$, by Anonymity.
- ▶ So Statewise Dominance says they're equally good. If the coin toss really is better, Statewise Dominance (also called the sure-thing principle) is not an appropriate assumption for the social welfare ordering.

The Objection to Ex Ante Pareto (1)

EQUAL CHANCES			EQUAL		
	Heads	Tails		Heads	Tails
Alex	10	0	Alex	5	5
Darius	0	10	Darius	5	5

- ▶ The other move. Ex Ante Pareto: equally good. Each person's expected wellbeing is 5 either way.
- ▶ But EQUAL CHANCES guarantees inequality: one person is sure to end up worse off than the other.

The Objection to Ex Ante Pareto (2)

EQUAL CHANCES			EQUAL		
	Heads	Tails		Heads	Tails
Alex	10	0	Alex	5	5
Darius	0	10	Darius	5	5

- ▶ We might use a lottery to “split” an indivisible good. But when the thing can be shared equally, people usually think the best thing is *to* share it.
- ▶ Intuitively, EQUAL CHANCES is at best an imperfect alternative to sharing.

Decide As You Would with Full Information

The Principle of Full Information

If you know that, whichever state obtains, you would rightly rank X above Y once you knew it, then rank X above Y now.

- ▶ If we knew the coin would land Heads, we'd be choosing between $(5, 5)$ and $(10, 0)$. The objector: sharing is better.
- ▶ If we knew it would land Tails, we'd be choosing between $(5, 5)$ and $(0, 10)$. Sharing is better again.
- ▶ So EQUAL is better than EQUAL CHANCES, whatever happens. But Ex Ante Pareto says they are equally good.

Equally Good, or Incomparable? (1)

- ▶ When each would get 10, the verdict ought to be symmetric.
Two options:
 - ▶ $(10, 0) \sim (0, 10)$: equally good. That's Anonymity.
 - ▶ $(10, 0)$ and $(0, 10)$ are *incomparable*: neither better, nor worse, nor equally good.
- ▶ If they're equally good, then $(11, 0) \succ (10, 0) \sim (0, 10)$: Alex getting 11 is better than Darius getting 10.

Equally Good, or Incomparable? (2)

- ▶ But that's not obvious. A big difference would obviously matter; a small one, not so obviously.
- ▶ Incomparability can survive small improvements: $(11, 0)$ might still be neither better nor worse than $(0, 10)$.
- ▶ Anonymity and Statewise Dominance work together: they give UNEQUAL $\sim$ EQUAL CHANCES.
- ▶ If that's the step we find implausible, we could reject Statewise Dominance — or Anonymity.

Rejecting Anonymity Instead of Statewise Dominance

- ▶ Rejecting Anonymity can't mean it's better for one of them to get the 10. So $(10, 0)$ and $(0, 10)$ are incomparable, and Statewise Dominance no longer gives $\text{UNEQUAL} \sim \text{EQUAL CHANCES}$.
- ▶ The intuition is that EQUAL CHANCES is better than either outcome for sure. So it must be better than both $(10, 0)$ and $(0, 10)$ —
- ▶ — even though, in each state of nature, its outcome is one of those two, and so is no better than them.

This Violates Negative Dominance

Negative Dominance

A prospect can't be better than another unless at least some outcome of the first is better than at least some outcome of the second.

- ▶ EQUAL CHANCES would be better than $(10, 0)$ for sure, though none of its outcomes is better than $(10, 0)$.
- ▶ Many people find Negative Dominance quite plausible.
- ▶ So perhaps one could avoid the theorem's conclusion by dropping Anonymity. But it's not so clear that that's a good way of doing it.

From Pairwise Anonymity to Anonymity (1)

Pairwise Anonymity

If outcome Y can be obtained from outcome X by swapping the wellbeing levels of two people, then $X \sim Y$.

$$(1, 2, 3) \sim (2, 1, 3) \sim (2, 3, 1)$$

$$(1, 2, 3) \sim (2, 3, 1)$$

By Transitivity.

From Pairwise Anonymity to Anonymity (2)

- ▶ Any rearrangement of n people can be made by swapping two at a time:
 - ▶ swap the level that should end up first into first place;
 - ▶ then the level that should end up second into second place;
 - ▶ and so on. At most $n - 1$ swaps.
- ▶ Each swap gives an equally good outcome, by Pairwise Anonymity.
- ▶ So, by Transitivity, any rearrangement is equally good: Anonymity.
- ▶ This works for any finite number of people. With infinitely many, it doesn't.

Ex Ante Pareto, One Person at a Time

- ▶ We only need Ex Ante Pareto for changes to one person's prospect: if only i 's prospect changes, and it's equally good for i , the result is equally good.

EQUAL

	Heads	Tails
Alex	5	5
Darius	5	5

By Transitivity: EQUAL CHANCES $\sim$ EQUAL. The same works in general: one person at a time.

Different Numbers of People

- ▶ What if the number of people can vary?

Neutral Addition

Adding people whose lives are neutral (wellbeing 0) leaves an outcome equally good.

- ▶ Add Neutral Addition to the theorem's premises, and we get total utilitarianism: $X \succsim Y$ iff X has at least as great total wellbeing as Y , whatever the numbers of people.
- ▶ Why: concentrate each outcome's wellbeing in one person, then remove the neutral lives. Next slide.
- ▶ Zero: 0 is the level where a life would be neither good nor bad. That fixes which lives are neutral.

Concentrate, Then Remove the Neutral Lives

A dash: that person doesn't exist.

	Alex	Darius	Philip	Total	Why
X	3	1	2	6	
$\sim$	6	0	0	6	The theorem
$\sim$	6	—	—	6	Neutral Addition
Y	2	5	—	7	
$\sim$	0	7	—	7	The theorem
$\sim$	—	7	—	7	Neutral Addition
$\sim$	7	—	—	7	Anonymity across different people

$(7, -, -) \succ (6, -, -)$: Alex at 7 is better than Alex at 6.

So, by Transitivity, $Y \succ X$: the greater total wins.

Optional Readings for Next Week

- ▶ Parfit, Derek (1997). “Equality and Priority”. In: *Ratio* 10.3, pp. 202–221. **I recommend this one most.**
- ▶ Temkin, Larry S. (2000). “Equality, Priority, and the Levelling Down Objection”. In: *The Ideal of Equality*. Ed. by Matthew Clayton and Andrew Williams. London/New York: Macmillan and St Martin’s Press, pp. 126–161.
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